



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



HelloKitty ransomware Malware Analysis

Malware Analysis Report

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TLP:CLEAR | Lost Boys Cyber | HelloKitty ransomware Malware Analysis - Malware Analysis Report

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HelloKitty Malware Analysis

1. Executive Summary

Lost Boys Cyber assesses the submitted Windows PE sample as HelloKitty ransomware with high confidence. The identification is supported by the embedded mutex `HelloKittyMutex`, a HelloKitty-style ransom note, the Tor payment portal `6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion`, the `.crypted` encrypted-file extension, and code-level evidence of recursive file enumeration, NTRU/AES-related encryption logic, WMI-based shadow-copy handling, and anti-recovery behavior.

The sample did not produce a usable guest process tree during the TRASHcan run, so dynamic coverage is limited. However, static and Ghidra headless reverse-engineering evidence is strong enough to classify the malware family and describe expected operational impact. If executed successfully in a victim environment, this sample is designed to encrypt files, modify file attributes, write a ransom note, remove or interfere with recovery paths, and direct the victim to a Tor negotiation portal.

Priority Finding	Evidence	Defensive Implication
HelloKitty ransomware identity	`HelloKittyMutex`, HelloKitty ransom note text, known HelloKitty onion portal, `.crypted` extension	Treat any host execution as a ransomware incident, not generic suspiciousware
File encryption workflow	Ghidra `FUN_00409a70` calls `nttru_crypto_nttru_encrypt`; imports include `ReadFile`, `WriteFile`, `MoveFileW`, `GetQueuedCompletionStatus`	Endpoint containment and rapid backup validation are urgent
Recovery interference	Ghidra `FUN_00409130` queries `Win32_ShadowCopy`; OSINT reports HelloKitty deletes shadow copies	VSS, backup, and recovery telemetry should be reviewed immediately
Dynamic telemetry gap	TRASHcan collected PCAP but no guest process tree, filesystem changes, registry changes, or memory dump	Dynamic non-observation should not be treated as benign behavior

2. Sample Metadata

Field	Value
Report title	HelloKitty Malware Analysis
Malware family / cluster	HelloKitty ransomware
Sample SHA256	`c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e`
Sample SHA1	`3294a12a583d2634f6e3d1232052dfe0cd51a44a`
Sample MD5	`8e4a887acab5f9475c5fa9a26fb9e720`
Submitted archive	`/home/lost0x01/internal-secops-portal/data/uploads/20260617_183629_893432/c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e.7z`
Archive password	`infected`

Extracted filename	`c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e`
File size	183,296 bytes
File type	PE32 executable, Intel 80386, Windows console subsystem
Architecture	x86 / 32-bit
Image base	`0x00400000`
Entry point	`0x00410783` from objdump PE header
Compile timestamp	Thu Nov 12 23:24:34 2020 UTC from PE header
Sections	`.text`, `.rdata`, `.data`, `.reloc`
Signing status	Unsigned in static triage
Sandbox verdict	Suspicious, score 35, confidence 0.35

3. Source Review & Confidence

Evidence Lane	Source	Result	Confidence
Submitted sample	Password-protected 7z archive supplied through portal task 122	One PE32 executable extracted and hashed	High
Static triage	`file`, `rabin2`, strings, capa, FLOSS, sandbox `static_triage.json`	Ransomware strings, HelloKitty mutex, Tor URL, encryption APIs, file traversal imports	High
Ghidra/code analysis	Ghidra 11.3.2 headless project `ghidra_proj/HelloKitty_c7`; output `evidence/ghidra_headless_keyr efs.txt`	811 functions analyzed; key functions and xrefs recovered	High
Dynamic analysis	TRASHcan sandbox run `2026-06-17_195710_c7d6719bbfb5`	PCAP collected; no guest process tree or filesystem/registry deltas recovered	Low to medium
OSINT	SentinelOne, Malwarebytes, Proven Data, Quorum Cyber search results	Public reporting associates this hash and onion portal with HelloKitty; behavior aligns with family reporting	Medium to high

Analyst confidence is high for family attribution and static/code behavior. Confidence is lower for runtime execution details because the sandbox artifacts explicitly state that no guest process tree, dropped files, registry changes, autoruns, or memory dump were collected.

4. Initial Triage and Delivery Assessment

The submitted artifact is a password-protected archive containing one Windows PE executable. The password `infected` and single-sample packaging are consistent with controlled malware submission rather than a native victim delivery container. No email lure, exploit document, script wrapper, or installer chain was provided in the evidence set.

Triage Signal	Observation	Assessment
Archive	7z, AES-encrypted, password `infected`	Malware-analysis safe-transfer format
Executable	PE32 console binary, x86	Native Windows payload
Timestamp	2020-11-12 23:24:34 UTC	Plausible for early HelloKitty-era samples; timestamp is not independently trusted
Packer	No clear packer signal; entropy 6.7049 in sandbox static findings	Not strongly packed, but contains crypto code and encoded data
Resource footprint	No resource directory in objdump data directories	Payload logic appears code/string driven rather than resource-staged
Initial verdict	Suspicious in TRASHcan, but statically ransomware-identifiable	Treat as malicious despite limited dynamic capture

5. Static Analysis

The binary exposes a conventional PE import table and visible ransomware strings. Static artifacts show the sample is not merely a dropper stub; it contains file traversal, encryption, WMI, mutex, command-line, and cleanup logic in the main executable.

Area	Evidence	Interpretation
Mutex	`HelloKittyMutex` at Ghidra address `00428c6c`; xrefs from `FUN_004095d0`	Single-instance execution and strong HelloKitty family marker
Ransom note	Embedded text begins `Hello dear user. Your files have been encrypted.`	User-facing extortion note generated by payload
Payment portal	`http://6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgr y5b44hhgfwc5ml5qrdad.onion`	Tor-based negotiation/support site associated with HelloKitty reporting
Extension	`.crypted` and `).crypted` strings at `004294cc` / `004294e0`	Encrypted-file marker
Recovery targeting	`select * from Win32_ShadowCopy` and `Win32_ShadowCopy.ID=%s`	WMI shadow-copy enumeration/deletion workflow
Crypto	`ntru_crypto_*`, AES-related capa hits, `CryptGenRandom`, `CryptAcquireContextW`	Hybrid encryption workflow using Windows PRNG and embedded NTRU/AES routines
File traversal	`FindFirstFileW`, `FindNextFileW`, `CreateFileW`, `ReadFile`, `WriteFile`, `MoveFileW`	Recursive file discovery and modification
Parallelism	`CreateThread`, `CreateIoCompletionPort`, `GetQueuedCompletionStatus`, `PostQueuedCompletionStatus`	Multi-threaded/IO completion-port file encryption design

Selected imports of analytical interest:

DLL	Imports	Relevance
`KERNEL32.dll`	`FindFirstFileW`, `FindNextFileW`,	File discovery, file modification,

	`CreateFileW`, `ReadFile`, `WriteFile`, `MoveFileW`, `SetFileAttributesW`, `CreateThread`, `CreateIoCompletionPort`	concurrency
`ADVAPI32.dll`	`CryptAcquireContextW`, `CryptGenRandom`, `CryptReleaseContext`	Random material generation for encryption
`OLE32.dll` / `OLEAUT32.dll`	`CoInitialize`, `CoCreateInstance`, `CoSetProxyBlanket`, COM helpers	WMI/COM interaction for system management tasks
`SHELL32.dll`	`CommandLineToArgvW`, `ShellExecuteW`, `SHEmptyRecycleBinA`	Argument parsing, command execution, recycle-bin cleanup
`SHLWAPI.dll`	`StrStrW`	File/path name filtering
`MPR.dll`	Network provider functions in import set	Supports network resource/share interaction

capa reported capabilities consistent with ransomware execution: recursive file enumeration, file write/read/move, file attribute modification, self-deletion through COMSPEC, mutex checks, process creation/termination, WMI namespace access through WbemLocator, network share discovery, system information discovery, AES constants, SHA1/SHA256 hashing, and random generation through WinAPI.

6. Code Analysis

GhidraMCP was reachable in the tool schema but returned an OpenCTI HTML page instead of a Ghidra program response, indicating the MCP endpoint was not connected to the active Ghidra project for this run. To preserve the required reversing lane, the sample was imported into Ghidra 11.3.2 headless using `analyzeHeadless`, producing project `ghidra\_proj/HelloKitty\_c7` and key-reference output at `evidence/ghidra\_headless\_keyrefs.txt`. Ghidra analyzed 811 functions.

Function / Address	Evidence	Analyst Interpretation
`FUN_004095d0` at `004095d0`	Opens/creates `HelloKittyMutex`; builds ransom note; references Tor URL; invokes command-line parsing and worker setup	Main orchestration routine
`FUN_00409130` at `00409130`	COM/WMI setup; connects to `ROOT\cimv2`; executes `select * from Win32_ShadowCopy`; formats `Win32_ShadowCopy.ID=%s`	Shadow-copy enumeration and likely deletion/recovery suppression routine
`FUN_00409a70` at `00409a70`	Uses IO completion status, `ReadFile`, `WriteFile`, crypto state marker `0xabccdcd`, and `ntru_crypto_ntru_encrypt`	Per-file encryption worker / read-write pool thread
`FUN_0040ab80` at `0040ab80`	Calls `FindFirstFileW` / `FindNextFileW`; skips selected names; clears read-only attributes with `SetFileAttributesW`; calls file-processing routine `FUN_0040a730`	Recursive directory traversal and file selection
`FUN_0040ff60` at `0040ff60`	Calls `CryptAcquireContextW`, `CryptGenRandom`, `CryptReleaseContext`	Windows cryptographic random material generation

Key decompiled evidence excerpts:

```
// FUN_004095d0: single-instance guard and ransom-note construction
DAT_0042dab0 = OpenMutexW(0x100000, 0, L"HelloKittyMutex");
if (DAT_0042dab0 != (HANDLE)0x0) {
    CloseHandle(DAT_0042dab0);
    ExitProcess(0);
}
DAT_0042dab0 = CreateMutexW((LPSECURITY_ATTRIBUTES)0x0, 0, L"HelloKittyMutex");
wsprintfW(note_buffer,
    L"Hello dear user... Your files have been encrypted... Open this website in tor browser: %s/%S",
    L"http://6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion");

// FUN_00409130: WMI shadow copy workflow
CoCreateInstance(...);
ConnectServer(L"R00T\\cimv2", ...);
CoSetProxyBlanket(...);
ExecQuery(..., L"select * from Win32_ShadowCopy", ...);
wsprintfW(local_13c, L"Win32_ShadowCopy.ID='%s'", local_2c);

// FUN_00409a70: encryption worker evidence
BVar6 = GetQueuedCompletionStatus(DAT_0042dac4, &local_558, &local_4c4, &local_564, 5000);
BVar6 = ReadFile(file_handle, buffer, 0x78000, 0, overlapped);
EnterCriticalSection(&DAT_0042cf78);
iVar7 = ntru_crypto_ntru_encrypt(DAT_0042cf74, key_len, key_blob, 0x48, public_key, out_len, out_buf);
LeaveCriticalSection(&DAT_0042cf78);

// FUN_0040ff60: PRNG helper
BVar2 = CryptAcquireContextW((HCRYPTPROV *)&param_4, 0, 0, 1, 0xf0000000);
BVar2 = CryptGenRandom((HCRYPTPROV)param_4, param_3, param_2);
CryptReleaseContext((HCRYPTPROV)param_4, 0);
```

The code layout supports a hybrid ransomware design: generate or derive per-run/per-file cryptographic material, use symmetric encryption for file content, protect key material with NTRU public-key encryption, append/rename encrypted files with `.crypted`, and write a ransom note directing the victim to the embedded Tor portal.

7. Dynamic Analysis

TRASHcan executed the sample in `win-malware-lab` from snapshot `clean-guestadditions-sysmon` using host-only gateway/DNS `192.168.56.1`; guest IP was `192.168.56.20`. The run captured `capture.pcapng` over approximately 104 seconds and produced static, network, and report artifacts. It did not collect a guest process tree or endpoint filesystem/registry changes.

Dynamic Area	Observation	Assessment
Process tree	`dynamic/process_tree.md`: no guest process tree collected; `process_count: 0`	Instrumentation gap or execution failure; cannot use absence as benign evidence
Filesystem changes	`dynamic_analysis.json`: none observed	Not reliable due to absent guest artifacts
Registry changes	none observed	Not reliable due to absent guest artifacts
Autoruns/persistence	none observed	Consistent with ransomware that may not require persistence, but coverage is limited
Dropped files	none observed	Ransom note/dropped artifacts were not captured dynamically
Network	PCAP present, 1,803 packets, 317 KB	Mostly Windows/Microsoft background traffic; no C2 candidate scored
Memory	memory dump not present; Volatility unavailable in artifact	No memory-based validation possible

The sandbox dynamic verdict was `suspicious`, with scoring reasons `suspicious static strings or commands` and `ATT&CK techniques mapped`. Because the sample contains direct code-level ransomware behavior, the dynamic non-observations should be written as coverage limitations, not exculpatory findings.

8. Process Tree and Execution Chain

No guest process tree was collected by TRASHcan. The expected execution chain from static/code evidence is:

Stage	Expected Behavior	Supporting Evidence
Launch	PE32 console executable starts and initializes COM	Imports <code>CoInitialize</code> , command-line parsing via <code>CommandLineToArgvW</code>
Single-instance guard	Check/create <code>HelloKittyMutex</code> ; exit if already running	Ghidra <code>FUN_004095d0</code> xrefs to <code>OpenMutexW</code> and <code>CreateMutexW</code>
Recovery handling	Enumerate shadow copies through WMI	Ghidra <code>FUN_00409130</code> query <code>select * from Win32_ShadowCopy</code>
File discovery	Recursively traverse directories and filter excluded names	<code>FUN_0040ab80</code> , <code>FindFirstFileW</code> , <code>FindNextFileW</code> , string exclusions such as <code>desktop.ini</code> , <code>autorun.inf</code> , <code>boot.ini</code> , <code>ntuser.dat</code>
Encryption workers	Use IO completion queues and worker threads to process files	<code>GetQueuedCompletionStatus</code> , <code>CreateThread</code> , <code>ReadFile</code> , <code>WriteFile</code> , <code>ntru_crypto_ntru_encrypt</code>
Ransom note / extension	Write ransom note and append <code>.crypted</code> marker	Embedded note and extension strings
Optional cleanup	Self-deletion command string <code>/C ping 127.0.0.1 & del %s</code>	Static string and capa self-delete capability

9. Persistence and System Modification

This sample does not show strong persistence behavior in the provided artifacts. That is typical for many ransomware payloads: the objective is rapid impact, not long-term foothold retention. System modification evidence is focused on encryption and recovery suppression.

Modification Type	Evidence	Defensive Action
File content modification	<code>ReadFile</code> / <code>WriteFile</code> ; <code>ntru_crypto_ntru_encrypt</code> ; <code>.crypted</code> extension	Hunt for mass write/rename activity and extension changes
File attribute modification	<code>SetFileAttributesW</code> clears read-only attributes before processing	Alert on read-only attribute clearing followed by writes/renames
Shadow copy interaction	WMI <code>Win32_ShadowCopy</code> query	Monitor WMI shadow-copy deletion/enumeration from unusual processes
Recycle-bin cleanup	<code>SHEmptyRecycleBinA</code> import and capa recycle-bin capability	Review suspicious recycle-bin clearing around ransomware alerts
Self-delete	<code>/C ping 127.0.0.1 & del %s</code> string and capa self-delete capability	Preserve volatile evidence quickly; malware may remove itself

No autorun keys, scheduled tasks, services, or startup-folder changes were reported by TRASHcan. That absence is a sandbox coverage statement, not a guarantee that related intrusion tooling did not establish persistence before ransomware deployment.

10. Network and Infrastructure Analysis

The only malware-relevant network indicator embedded in the sample is the Tor negotiation URL. The sandbox PCAP mainly shows routine Windows background connectivity and multicast/name-resolution traffic. No live C2 candidate, fake-service hit, Suricata alert, or HTTP callback to the onion URL was observed.

Indicator	Type	Source	Assessment
`6x7dp6h3w6q3ugjv4yv5gyjc3femb24kysgry5b44hhgfwc5ml5qrdad.onion`	Tor hidden service	Static strings, Ghidra xref, OSINT	HelloKitty ransom/payment portal; block/detect Tor access attempts where policy permits
`http://6x7dp6h3w6q3ugjv4yv5gyjc3femb24kysgry5b44hhgfwc5ml5qrdad.onion`	URL	Static strings	Embedded ransom-note contact URL
`ctldl.windowsupdate.com`, `www.msftconnecttest.com`, `settings-win.data.microsoft.com`, etc.	DNS/HTTP/TLS	Sandbox PCAP	Windows background traffic; not attributed to malware C2 in this run
`hxmredkic`, `stlpuvlipv`, `udoakoqt` and `.local` variants	Local DNS/name queries	Sandbox PCAP	Host/lab local-name artifacts; no external C2 linkage established

SentinelOne reporting notes the same onion address in HelloKitty ransom notes and states that HelloKitty variants may use AES/RSA or NTRU/AES encryption recipes. Malwarebytes reporting lists this exact SHA256 among HelloKitty IOCs and describes `.crypted` / `.kitty` extensions, process/service termination, and shadow-copy deletion as family behaviors.

11. Indicators of Compromise

Primary IOCs suitable for blocking, triage, or retro-hunt are below. Several sandbox-generated IOC rows include benign Microsoft connectivity and file names misclassified as domains; those should not be promoted to blocklists without analyst review.

Type	Value	Source	Confidence	Notes
SHA256	`c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e`	Submitted sample	High	Primary sample
SHA1	`3294a12a583d2634f6e3d1232052dfe0cd51a44a`	Local hashing	High	Primary sample
MD5	`8e4a887acab5f9475c5fa9a26fb9e720`	Local hashing	High	Primary sample
Mutex	`HelloKittyMutex`	Static/Ghidra	High	Family marker
URL	`http://6x7dp6h3w6q3ugjv4yv5gyjc3femb24kysgry5b44hhgfwc5ml5qrdad.onion`	Static/Ghidra	High	Ransom portal
Domain	`6x7dp6h3w6q3ugjv4yv5gyjc3femb24kysgry5b4`	Static/Ghidra/OSINT	High	Tor hidden service

	4hhgfwc5ml5qrdad.onion`			
File extension	`.crypted`	Static/Ghidra	High	Encrypted-file marker
Ransom note text	`Hello dear user. Your files have been encrypted.`	Static/Ghidra	High	User-facing note content
WMI query	`select * from Win32_ShadowCopy`	Static/Ghidra	High	Recovery suppression behavior
Self-delete command fragment	`/C ping 127.0.0.1 & del %s`	Static strings	Medium	Delayed self-delete pattern
Exclusion-like filenames	`desktop.ini`, `autorun.inf`, `boot.ini`, `ntuser.dat`, `pagefile.sys`, `swapfile.sys`	Static strings	Medium	File-selection logic; not malicious alone

Recommended normalized IOC sidecar fields for downstream systems:

```
type,value,confidence,context
sha256,c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e,high,HelloKitty ransomware sample
sha1,3294a12a583d2634f6e3d1232052dfe0cd51a44a,high,HelloKitty ransomware sample
md5,8e4a887acab5f9475c5fa9a26fb9e720,high,HelloKitty ransomware sample
mutex,HelloKittyMutex,high,Single-instance family marker
url,http://6x7dp6h3w6q3ugjv4yv5gydj3femb24kysgry5b44hhgfwc5ml5qrdad.onion,high,Embedded ransom portal
file_extension,`.crypted`,medium,Encrypted output extension
```

12. MITRE ATT&CK Mapping

Tactic	Technique	Evidence	Confidence
Execution	T1059 - Command and Scripting Interpreter	Static self-delete command uses `cmd.exe /C`; capa mapped command interpreter	Medium
Execution	T1047 - Windows Management Instrumentation	Ghidra `FUN_00409130` connects to `ROOT\cimv2` and queries `Win32_ShadowCopy`	High
Execution	T1129 - Shared Modules	capa identified shared module usage/loading	Medium
Defense Evasion	T1027 - Obfuscated Files or Information	capa identified encoded/Base64/XOR-related behavior and crypto constants	Medium
Defense Evasion	T1070.004 - Indicator Removal: File Deletion	Self-delete string `/C ping 127.0.0.1 & del %s`; capa self-delete	Medium
Defense Evasion	T1222 - File and Directory Permissions Modification	`SetFileAttributesW` clears read-only attributes before processing	High
Discovery	T1083 - File and Directory	`FindFirstFileW` /	High

	Discovery	`FindNextFileW`; recursive traversal function	
Discovery	T1082 - System Information Discovery	`GetSystemInfo` import and capa system information discovery	Medium
Discovery	T1135 - Network Share Discovery	capa network share discovery; MPR import set	Medium
Discovery	T1614.001 - System Language Discovery	capa keyboard/system language discovery via `GetKeyboardLayoutList`	Medium
Impact	T1486 - Data Encrypted for Impact	Ransom note, `.crypted`, NTRU/AES evidence, file read/write encryption loop	High
Impact	T1490 - Inhibit System Recovery	WMI `Win32_ShadowCopy` workflow and family OSINT	High

13. Detection Engineering

Detection should combine high-confidence static family markers with behavior-based ransomware analytics. Hash-only detections are useful for this sample but insufficient for variants.

Detection	Data Source	Purpose	Priority
HelloKitty static YARA	File scanning / EDR file telemetry	Identify this family/sample lineage by mutex, note, Tor URL, extension, and NTRU strings	High
Shadow-copy WMI query	Sysmon Event ID 1/7/11/22, EDR process telemetry, PowerShell/WMI logs	Detect ransomware recovery suppression	High
Mass `.crypted` writes	EDR file events	Detect active encryption impact	Critical
Tor ransom URL access	Proxy/DNS/firewall/Tor gateway telemetry	Detect victim negotiation or operator validation activity	Medium
IOCP + file rewrite burst from unsigned PE	EDR process/file telemetry	Behavioral ransomware detection when static markers change	High

YARA rule:

```
rule LBC_HelloKitty_Ransomware_c7d6719b
{
  meta:
    description = "Detects HelloKitty ransomware sample/cluster with HelloKittyMutex, ransom note, Tor portal, .crypted extension, and NTRU strings"
    author = "Lost Boys Cyber"
    reference_sha256 = "c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e"
    date = "2026-06-17"
  strings:
    $mz = { 4D 5A }
    $mutex_w = "HelloKittyMutex" wide ascii
    $note_w = "Your files have been encrypted" wide ascii
```

```

$onion_a = "6x7dp6h3w6q3ugjv4yv5gydj3femb24kysgry5b44hhgfwc5ml5qrdad.onion" ascii wide
$ext_w = ".crypted" wide ascii
$wmi_w = "select * from Win32_ShadowCopy" wide ascii
$ntru1 = "ntru_crypto_ntru_encrypt" ascii
$ntru2 = "ntru_crypto_drbg_generate" ascii
condition:
  uint16(0) == 0x5a4d and
  4 of ($mutex_w, $note_w, $onion_a, $ext_w, $wmi_w, $ntru*)
}

```

Sigma rule for shadow-copy WMI behavior:

```

title: HelloKitty-Style Shadow Copy WMI Query From Untrusted Process
id: 5dfb08e8-9eb4-47f7-a90f-hellokitty-shadowcopy
status: experimental
description: Detects processes querying Win32_ShadowCopy in a pattern consistent with ransomware
recovery suppression.
author: Lost Boys Cyber
date: 2026/06/17
logsource:
  product: windows
  category: process_creation
detection:
  selection_cmdline:
    CommandLine|contains:
      - 'Win32_ShadowCopy'
      - 'select * from Win32_ShadowCopy'
  filter_admin_tools:
    Image|endswith:
      - '\\wbem\\WMIC.exe'
      - '\\WindowsPowerShell\\v1.0\\powershell.exe'
  condition: selection_cmdline
fields:
  - Image
  - CommandLine
  - ParentImage
  - User
  - Computer
falsepositives:
  - Backup administration
  - EDR or inventory tooling
level: high
tags:
  - attack.impact
  - attack.t1490
  - attack.execution
  - attack.t1047

```

Microsoft Defender XDR / KQL hunt for `.crypted` bursts:

```

DeviceFileEvents
| where Timestamp > ago(30d)
| where FileName endswith ".crypted"
| summarize
  EncryptedFileCount=count(),
  FirstSeen=min(Timestamp),
  LastSeen=max(Timestamp),
  SamplePaths=make_set(FolderPath, 10)
by DeviceName, InitiatingProcessFileName, InitiatingProcessSHA256, InitiatingProcessCommandLine
| where EncryptedFileCount >= 10
| order by EncryptedFileCount desc

```

KQL hunt for the sample hash or mutex-adjacent process:

```

let HelloKittyHashes = dynamic(["c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e"]);
DeviceProcessEvents
| where Timestamp > ago(90d)
| where SHA256 in (HelloKittyHashes)
  or ProcessCommandLine has_any ("HelloKittyMutex",

```

```
"6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion")
| project Timestamp, DeviceName, FileName, SHA256, ProcessCommandLine, InitiatingProcessFileName,
AccountName
```

Splunk hunt for recovery suppression plus encryption extension:

```
(index=win OR index=edr)
((CommandLine="*Win32_ShadowCopy*" OR process_command_line="*Win32_ShadowCopy*")
OR (file_name="*.crypted" OR TargetFilename="*.crypted"))
| stats count min(_time) as firstTime max(_time) as lastTime values(CommandLine) as cmd
values(TargetFilename) as files by host user process_name parent_process_name process_hash
| where count > 5 OR mvcount(files) > 5
```

Suricata-style HTTP/Tor URL content detection for proxy logs or decrypted HTTP telemetry:

```
alert http any any -> any any (msg:"LBC HelloKitty ransom onion URL in HTTP traffic"; http.host;
content:"6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion"; nocase; classtype:trojan-
activity; sid:42067191; rev:1;)
```

14. Threat Hunting

Threat hunts should prioritize impact and recovery-suppression signals because the sandbox did not provide a full process tree. Run hunts across at least 90 days of EDR, file, WMI, DNS/proxy, and backup telemetry.

Hypothesis	Query / Signal	Expected Finding
HelloKitty executed in the environment	Sample hash, mutex string, ransom URL, `crypted` writes	Patient-zero host and execution time
Ransomware attempted recovery suppression	WMI queries for `Win32_ShadowCopy`, VSS deletion, backup-service interference	Evidence of impact preparation
Encryption occurred at scale	Burst of file renames/writes ending `crypted`	Affected hosts/shares and blast radius
Operator used Tor negotiation portal	DNS/proxy/Tor gateway access for the onion address	Victim/operator contact attempts
Pre-ransomware intrusion existed	Remote admin tools, credential access, SMB/RDP fan-out before encryption	Initial access and lateral movement context

KQL for WMI shadow copy activity:

```
DeviceProcessEvents
| where Timestamp > ago(90d)
| where ProcessCommandLine has "Win32_ShadowCopy"
  or ProcessCommandLine has_all ("shadow", "delete")
  or ProcessCommandLine has "vssadmin delete shadows"
| project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine, InitiatingProcessFileName,
InitiatingProcessCommandLine, SHA256
| order by Timestamp desc
```

KQL for file-attribute clearing followed by encrypted extensions:

```
let cryptedWrites = DeviceFileEvents
| where Timestamp > ago(30d)
| where ActionType in ("FileCreated", "FileModified", "FileRenamed")
| where FileName endswith ".crypted"
| project DeviceId, DeviceName, Timestamp, FolderPath, InitiatingProcessSHA256,
InitiatingProcessFileName;
cryptedWrites
| summarize Count=count(), FirstSeen=min(Timestamp), LastSeen=max(Timestamp),
Paths=make_set(FolderPath, 25)
  by DeviceName, InitiatingProcessFileName, InitiatingProcessSHA256
| where Count > 5
| order by Count desc
```

Splunk for suspicious Tor/onion references:

```
(index=proxy OR index=dns OR index=edr)
("6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion" OR "HelloKittyMutex" OR "Your files
have been encrypted")
| stats count min(_time) as firstTime max(_time) as lastTime values(src_ip) as src values(user) as user
values(process_name) as process by host sourcetype
```

KQL for known hash and adjacent execution:

```
let h = "c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e";
DeviceProcessEvents
| where Timestamp > ago(180d)
| where SHA256 == h or InitiatingProcessSHA256 == h
| join kind=leftouter (
    DeviceFileEvents
    | where Timestamp > ago(180d)
    | where InitiatingProcessSHA256 == h
    | summarize FilesTouched=count(), SampleFiles=make_set(FolderPath, 20) by DeviceId
) on DeviceId
| project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine, SHA256, FilesTouched,
SampleFiles
```

15. Triage and Incident Response

If this sample is detected or suspected in an environment, treat the event as a ransomware emergency.

Phase	Action	Rationale
Immediate containment	Isolate affected endpoints and disable SMB/share access for suspected hosts	Prevent continued encryption and share traversal
Evidence preservation	Capture EDR triage package, volatile process/network state, ransom notes, encrypted-file examples, and suspicious executables	The sample may self-delete and dynamic artifacts are time-sensitive
Scope	Search for the hash, `.crypted` extension, ransom note text, Tor URL, and WMI shadow-copy activity	Establish affected hosts and encryption window
Recovery	Validate offline/immutable backups before restore; inspect VSS and backup logs for deletion attempts	HelloKitty family behavior targets recovery paths
Credential hygiene	Rotate privileged and service-account credentials exposed before encryption	Ransomware deployment is often the final phase of a broader intrusion
Communications	Preserve ransom note and onion URL for legal/IR coordination; do not browse from production systems	Avoid contaminating evidence or exposing negotiator identity

Recommended first-hour checklist:

- Quarantine the endpoint from network shares and remote management paths.
- Export EDR telemetry around process creation, file modifications, WMI, and network traffic.
- Search for `HelloKittyMutex`, `.crypted`, and the onion address across EDR, SIEM, proxy, and DNS logs.
- Confirm backup integrity from known-good points before any restore attempt.
- Identify whether any shadow copies were enumerated or deleted.
- Preserve the submitted binary and encrypted-file samples with hashes and chain-of-custody notes.

16. Validation and Emulation

Validation should be performed in an isolated lab only. The provided TRASHcan run restored the VM snapshot after execution and captured a PCAP, but did not capture endpoint process or filesystem telemetry. Future validation should focus on endpoint instrumentation rather than re-running the sample without controls.

Validation Goal	Safe Method	Expected Evidence
Static family detection	Run YARA against extracted sample only	Match on mutex, ransom note, Tor URL, `.crypted`, NTRU strings
File-impact behavior	Use a disposable VM with synthetic documents and full Sysmon/EDR collection	Process tree, file writes/renames, ransom note creation
Recovery suppression	Enable WMI/Sysmon logging for shadow-copy access	`Win32_ShadowCopy` query or deletion method
Network behavior	Capture traffic with DNS/HTTP/Tor egress controls	No direct C2 expected except ransom-note onion reference unless victim browses manually
Detection QA	Replay generated telemetry into SIEM where possible	KQL/SPL/Sigma alerts fire on `.crypted` and WMI/shadow-copy signals

Do not execute the sample on production endpoints, analyst workstations, or shared file systems. Use disposable snapshots, host-only networking, fake services, and synthetic files. Because this sample contains working encryption logic, any detonation environment should be assumed destructive.

17. Recommendations

Priority	Recommendation	Owner	Validation
Critical	Block or alert on the sample hash and high-confidence YARA rule in EDR/file-scanning platforms	SecOps	Confirm detections fire on the extracted sample in a safe repository
Critical	Hunt for `.crypted` file creation/rename bursts and recover affected hosts from immutable backups	IR / Infrastructure	Produce host/share list with first/last encrypted file timestamp
High	Monitor WMI `Win32_ShadowCopy` queries and shadow-copy deletion commands from non-admin tooling	Detection Engineering	Sigma/KQL detections return expected test events
High	Restrict Tor access and alert on the embedded onion address in proxy/DNS/security gateway logs	Network Security	Proxy/SIEM alert triggers on test indicator lookup
High	Harden and monitor backup systems, including backup-service credentials and administrative consoles	Infrastructure	Backup restore test passes from offline/immutable source
Medium	Review exposed	IR / IAM	Timeline shows or rules out

	RDP/VPN/remote-admin access and privileged account activity preceding any ransomware alert		initial-access and lateral movement paths
Medium	Add file-server canary files and high-volume rename/write alerts for ransomware early warning	Detection Engineering	Alerts fire during controlled rename/write simulation
Medium	Retain malware, Ghidra notes, IOCs, and detections in the analyst-kit source bundle for future enrichment	Threat Intel	Source bundle contains report, IOCs, detections, and evidence notes

18. References and Refresh Notes

- SentinelOne, `HelloKitty Ransomware Lacks Stealth, But Still Strikes Home`, March 2021. Reporting describes HelloKitty emergence in late 2020, association with CD Projekt Red, service/process termination behavior, NTRU/AES encryption variants, `.crypted` extension, and the same onion portal observed in this sample.
- Malwarebytes, `HelloKitty: When Cyberpunk met cy-purr-crime`, March 2021. Reporting states HelloKitty is named after `HelloKittyMutex`, lists this sample SHA256 as an IOC, describes `.kitty` / `.crypted` extensions, shadow-copy deletion, and Tor ransom-note URLs.
- Proven Data, `Decrypting HelloKitty Ransomware`, search result listing includes SHA256 `c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e` with 183,296-byte EXE metadata.
- Quorum Cyber, `Threat Intelligence: Vice Society Ransomware Report`, search result lists the same SHA256 among ransomware-related IOCs; this should be treated as enrichment context, not proof that the submitted sample was deployed by Vice Society in the incident under review.
- TRASHcan local sandbox artifacts, `/home/lost0x01/internal-secops-portal/data/sandbox-runs/2026-06-17\_195710\_c7d6719bbfb5/`, including `analysis.md`, `summary.json`, `static\_triage.json`, `dynamic/dynamic\_analysis.json`, `network/network\_summary.json`, `detections/iocs\_full.csv`, and `rule.yar`.
- Ghidra headless reversing output, `/home/lost0x01/.hermes/kanban/workspaces/t\_b0d0f411/evidence/ghidra\_headless\_keyrefs.txt`, generated from Ghidra 11.3.2 project `/home/lost0x01/.hermes/kanban/workspaces/t\_b0d0f411/ghidra\_proj/HelloKitty\_c7`.

Refresh notes:

- Re-check public reporting and malware repositories before publication if the final report date moves materially beyond 2026-06-17.
- Validate detections in the target SIEM/EDR before production deployment; field names differ between Microsoft Defender, Splunk CIM, Sysmon, and vendor EDR schemas.
- Treat the dynamic-analysis section as limited coverage unless a future run captures guest process/file/registry telemetry successfully.